Anway Pimpalkar

PhD Student and NDSEG Fellow at Harvard University

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EDUCATION

Harvard University 2025 –
PhD, Engineering Sciences: Materials Science & Mechanical Engineering
Secondary Field in Mind, Brain and Behaviour
Advisor: Katia Bertoldi, PhD

Johns Hopkins University 2023 – 2025
MSE, Biomedical Engineering
Thesis: Bridging Perception & Performance: Generating, Calibrating, & Evaluating Multimodal Sensory Feedback
Advisor: Jeremy D. Brown, PhD

College of Engineering Pune, India 2019 – 2023
BTech, Electronics and Telecommunication Engineering

AWARDS & GRANTS

Prize Fellowship • Harvard University, School of Engineering and Applied Sciences 2025 – 2031
$281,796 • 6 yr tuition, 2 yr stipend

National Defense Science & Engineering Graduate Fellowship • US DOW, Office of Naval Research 2025 – 2028
$277,484 • 3 yr tuition, 3 yr stipend

GEM Associate Fellowship • National GEM Consortium 2025 – 2031

Best Student Paper Finalist [for C2] • International Conference on Rehabilitation Robotics (ICORR) 2025

NIA Start-Up Challenge Finalist • National Institute on Aging of the NIH 2024 – 2025
$10,000 • Article ↗

Best Paper Award [for C1] • International Conference on Biomedical Engineering Science and Tech. 2023

Mitacs Globalink Research Award • Dalhousie University, Canada 2022
CA $9,915

Best Project Award [for O1] • IEEE National Project Competition, India 2021
₹2,000

RESEARCH EXPERIENCE

Harvard University • *Bertoldi Lab* ↗ Cambridge, MA
Doctoral Research Fellow • Advisor: Katia Bertoldi, PhD Oct 2025 –

- Multistable mechanisms and adaptive structures for interactive human-device interfaces.
- Mechanical systems and assistive robotics to support cognitive and physical functions.

Johns Hopkins University • *Haptics and Medical Robotics Lab* ↗ Baltimore, MD
Graduate Research Assistant • Advisor: Jeremy D. Brown, PhD Jan 2024 – Jun 2025

- Multimodal visual-haptic neurorehabilitation for improving post-stroke finger dexterity. [J4, C2, C3, C5, A6, A7]
- Soft multimodal haptic interfaces. [J3, A3, A4]

Harvard University • *Biohybrid Organs and Neuroprosthetics Lab* ↗ Cambridge, MA
Research Fellow • Advisor: Shriya S. Srinivasan, PhD May – Aug 2024

- Peripheral nerve stimulation to modulate cancer tumor metastasis.
- Surgical approach for treating stress urinary incontinence.

IIT Bombay • *Human Motor Neurophysiology and Neuromodulation Lab* ↗ Mumbai, India
Research Intern • Advisor: Nivethida Thirugnanasambandam, PhD Jan – Aug 2023

- Quantification of the prospective component of sense of agency using machine learning.

College of Engineering Pune • *Department of Applied Sciences* ↗ Pune, India
Undergraduate Researcher • Advisor: Mahesh H. Shindikar, PhD Aug 2022 – Aug 2023

- Deep learning-based MRI skull tissue segmentation. [C1, A1]

Dalhousie University • *Agricultural Mechanized Systems Group* ↗ Truro, Canada
Mitacs Globalink Research Intern • Advisor: Travis J. Esau, PhD May – Aug 2022

- Automation of wild blueberry *(Vaccinium angustifolium)* harvesters to reduce operator cognitive load.

Queliz Lifetech • *Early Stage Medtech Startup* Pune, India
Research Intern, Electronics Subsystem Jun – Sep 2021

- Point-of-care device for finger dexterity rehabilitation in acute burn patients.

JOURNAL PUBLICATIONS

J4 **Pimpalkar A**, Brown JD. *Pneumatactors: Soft 3D-printed interface for co-located transient and continuous tactile stimuli.* In preparation. 2026.

J3 **Pimpalkar A**, West AM, Rai D, Xu J, Brown JD. *Perceptually calibrated haptic feedback provides limited benefit beyond vision in a virtual precision grip task.* In review. 2026.

J2 Harris C, **Pimpalkar A**, Aggarwal A, Yang J, Chen X, Schmidgall S, Rapuri S, Greenstein JL, Taylor CO, Stevens RD. *Preoperative risk prediction of major cardiovascular events in noncardiac surgery using the 12-lead electrocardiogram: explainable deep learning approach.* British Journal of Anesthesia. 2025 Sep; doi: 10.1016/j.bja.2025.07.085 ↗

Featured in *The Hub at Johns Hopkins*. Article ↗

J1 **Pimpalkar A**, Slepyan A, Thakor NV. *Vibrations at first contact encode object stiffness before grasp completion.* IEEE Sensors Letters. 2025 Aug; 9(8):1-4. doi: 10.1109/lsens.2025.3584860 ↗

Featured in *New Scientist Magazine*. Article ↗

CONFERENCE PRESENTATIONS † presenter

FULL-LENGTH PAPERS

C5 West AM, Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. *Cross-modal matching is not bidirectional: Implications for human multisensory feedback*. IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob), 2026 Aug. In print. [Oral Pres]

C4 Salgado AS, Bartlett JS, **Pimpalkar A**, Brown JD, Reed KB †. *Teaching at the Teachable Moment Using Asynchronous Videos Combined with Live Interactions.* ASEE Annual Conference & Exposition. 2026 Jun. In print. [Oral Pres]

C3 West AM †, **Pimpalkar A**, Xu J, Brown JD. *SenseMatch: Smartphone-based cross-modal matching for accessible perceptual assessments.* International IEEE/EMBS Conference on Neural Engineering (NER), 2025 Nov. p. 85-90. doi: 10.1109/ner61569.2025.11589033 ↗ [Spotlight Poster Pres]

C2 **Pimpalkar A**†, West AM, Xu J, Brown JD. *Optimizing cross-modal matching for multimodal motor rehabilitation.* International Conference On Rehabilitation Robotics (ICORR). 2025 May. P. 559-566. doi: 10.1109/icorr66766.2025.11063112 ↗ [Podium Pres] Best Student Paper Award Finalist.

C1 **Pimpalkar A** †, Patole RK, Kamble KD, Shindikar MH. *Performance evaluation of vanilla, residual, and dense 2D U-Net architectures for skull stripping of augmented 3D T1-weighted MRI head scans.* International Conference on Biomedical Engineering Science and Technology. 2023 Feb. p. 131-142. doi: 10.1007/978-3-031-54547-4_11 ↗ [Podium Pres] Best Paper Award.

ABSTRACTS AND SHORT PAPERS

A7 West AM †, Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. *Cross-modal matching is not bidirectionally invertible: Implications for multisensory representation*. 2026 Neural Control of Movement Meet. 2026 Apr. [Poster Pres]

A6 **Pimpalkar A** †, Rai D, Bartels UJ, Xu J, Brown JD. *Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task.* 2024 Biomedical Engineering Society Annual Meeting (BMES). 2024 Oct. [Podium Pres]

A5 Song H † et al. [including **Pimpalkar A**]. *Towards visual function restoration through photoacoustic stimulation.* Investigative Ophthalmology & Visual Science ARVO Annual Meeting. 2024 Jun. [Podium Pres]

A4 **Pimpalkar A** †, Ameta P †, Dalia A, Brown JD. *Pneumatactor arrays for high frequency vibrotactile feedback.* IEEE Haptics Symposium. 2024 Apr. [Live Demo, Work-in-Prog Paper]

A3 **Pimpalkar A** †, Ameta P †, Dalia A †, Brown JD, *Pneumatactor arrays for high frequency vibrotactile feedback.* US Senate AI Caucus Robotics Demo Day. 2024 Apr. [Live Demo]

A2 Harris C †, **Pimpalkar A**, Aggarwal A, Yang P, Chen X, Taylor CO, Greenstein J, Stevens RD. *Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms.* Johns Hopkins Medicine & Engineering Research Retreat. 2024 Feb. [Poster Pres]

A1 **Pimpalkar A** †, Patole R, Kamble K, Shindikar M. *Evaluating U-Nets for skull stripping of augmented T1-weighted MRI scans.* No Garland Neuroscience Conference. 2023 Feb. [Podium Pres]

NON PEER-REVIEWED ARTICLES

O1 **Pimpalkar A**, Niture D. *Towards contactless elevators with tinyML using person detection and keyword spotting.* 2024 Jul. doi: arXiv:2405.13051 ↗ Best Project Award.

TEACHING ENGAGEMENTS

Principles of the Design of Biomedical Instrumentation • Johns Hopkins University 580.771 Fall 2024

Teaching Assistant to Nitish V. Thakor, PhD

- Mentored labs on designing ECG, EMG, EEG circuits and projects including accessible gaming, wearables.

Haptic Interface Design for Human-Robot Interaction • Johns Hopkins University 530.691 Fall 2024

Class Researcher with Jeremy D. Brown, PhD

- Helped test an online learning platform providing self-paced lectures incorporating live instructor interaction. [C4]

Mechatronics • Johns Hopkins University 530.421 Spring 2024

Teaching Assistant to Jeremy D. Brown, PhD

- Built a scalable real-time location polling of autonomous air hockey robots using CV-based ArUco tag detection and ZigBee mesh networking. Led labs on building autonomous robots. GitHub ↗

MENTORING

UNDERGRADUATES

George Sealy • Summer Intern from University of Cambridge, Materials Science Jul – Sep 2026

Project: Mechanics and acoustics of elastic structures.

Jocelyn Hwang • Summer Intern from MIT, Electrical Engineering and Computer Science Jun – Sep 2026

Project: Multiplexing electronics for multi-electrode biosensing.

Fuxing (Fu) Chen • Undergraduate Researcher from MIT, Mechanical Engineering Jan 2026 –

Project: Elastic actuators for ingestible robots and drug delivery.

PRESENTATIONS AND INVITED LECTURES

T2 **Indian Institute of Technology Bombay** • Human Motor Neurophysiology & Neuromodulation Lab Jun 2025

T1 **Johns Hopkins University** • Neural & Rehabilitation Engineering Course Sep 2024

ACADEMIC SERVICE

IEEE Haptics Symposium • Student Volunteer 2024

Canadian Society of Bioengineering Annual Meeting • Student Volunteer 2022

IN MEDIA

Johns Hopkins University Hub ↗ Sep 17, 2025
"AI fares better than doctors at predicting deadly complications after surgery"

New Scientist Magazine ↗ Jan 2, 2025
"Delicate robot hands know just how hard to squeeze"

Johns Hopkins University Hub ↗ May 9, 2025
"A game-changing gait lab in a shoe"

Johns Hopkins University Hub ↗ May 16, 2024
"Robotics teams shoot it out in JHockey tournament"

Arduino Blog ↗ Jan 29, 2022
"This contactless system combines embedded ML and sensors to improve elevator safety"

LEADERSHIP AND TEAMWORK

Johns Hopkins India Institute • Operations Team 2024 – 2025
- Managed on-day operations and logistics for the first annual Hopkins-India Conference.

Johns Hopkins Biomedical Engineering • Master's Program Representative 2024 – 2025

COEP Impressions Cultural Festival • Head of Events (2021), Volunteer 2019 – 2021
- Led the planning and execution of 28 annual events with 10k attendees, managing a ₹2m budget and coordinating with an overall team of 126 members to execute concerts, competitions, and international celebrity engagements.
- Built and directly led a team of 15, working with 2 co-heads to build a platform from the ground up.

Rowing Team • COEP, Pune City • Competitive 1X, 2X, 4X Sculler 2020 – 2022
- Won race events and an institutional award commending skill and enthusiasm. Represented city at regional races.

COEP Mental Health & Wellness Center • Student Volunteer 2020 – 2022

COEP Data Science & AI Club • Research Lead (2020-21), Member 2020 – 2022